

Task 1

A file is encrypted using Veracrypt (A disk encryption tool). The password to access the file is encrypted in a hash format and provided to you in the drive with the name encoded.txt. Decode the password and enter in the vera crypt to unlock the file and find the secret code in it. The veracrypt setup file will be provided to you.

Attack Name: Password Recovery and Disk Encryption Analysis

Severity: Medium (CVSS Score: 5.9)

Reason:

Weak or commonly used passwords can often be recovered if their MD5 hashes are available in public hash databases. Once the password is recovered, encrypted containers protected with weak credentials can be accessed, compromising the confidentiality of sensitive data.

Steps to Reproduce:

Step 1: Open the Encoded File

The encoded.txt file was opened to view the encrypted password.

Hash Value: 482c811da5d5b4bc6d497ffa98491e38

Step 2: Identify the hash used

The hash value was analyzed using the hash-identifier tool, which identified it as an MD5 hash. This helped determine the appropriate method for password recovery.

```
(shravani@shravani)-[~]
$ hash-identifier 482c811da5d5b4bc6d497ffa98491e38
#####
#                                     #
#  ded.txt  %                               #
#  . Bin                                     #
#  ' ,                                         #
#  V V V V V V V V V V V V V V V V V V V V V V v1.2 #
#                                     By Zion3R #
#                                     www.Blackexploit.com #
#                                     Root@Blackexploit.com #
#####

Possible Hashs:
[+] MD5
[+] Domain Cached Credentials - MD4(MD4(($pass)).(strtolower($username)))

Least Possible Hashs:
[+] RAdmin v2.x
[+] NTLM
[+] MD4
[+] MD2
[+] MD5(HMAC)
[+] MD4(HMAC)
```

Fig 8: MD5 Hash Identification Using Hash Identifier

Step 3: Recover the password

An online MD5 hash lookup tool was used to recover the original password.

Recovered Password: password123

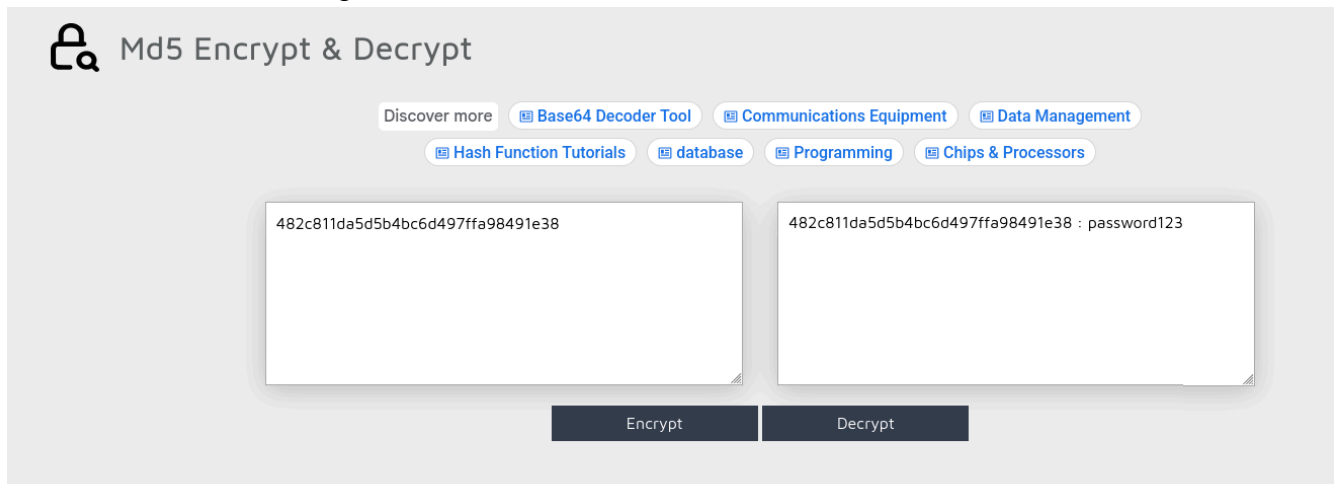


Fig 9: Password Hash Cracker

Step 4: Open VeraCrypt

VeraCrypt was launched, and the encrypted container file was selected. An available drive letter was chosen to prepare the volume for mounting using the recovered password.

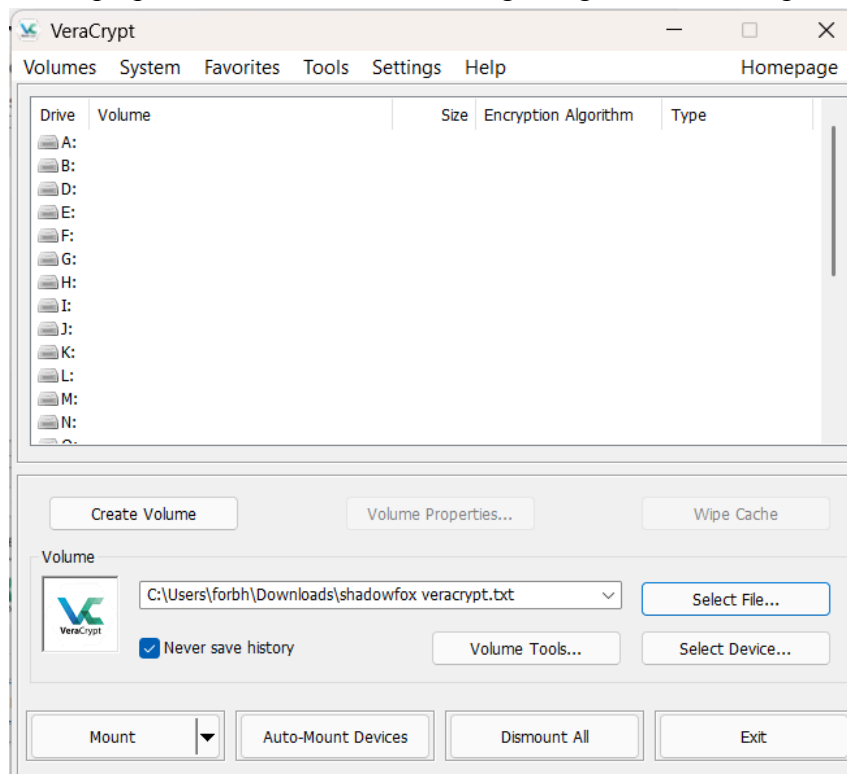


Fig 10: VeraCrypt Interface with Encrypted Container

Step 5: Enter the Recovered Password

The recovered password was entered into the VeraCrypt authentication window to verify access to the encrypted container before mounting the volume.

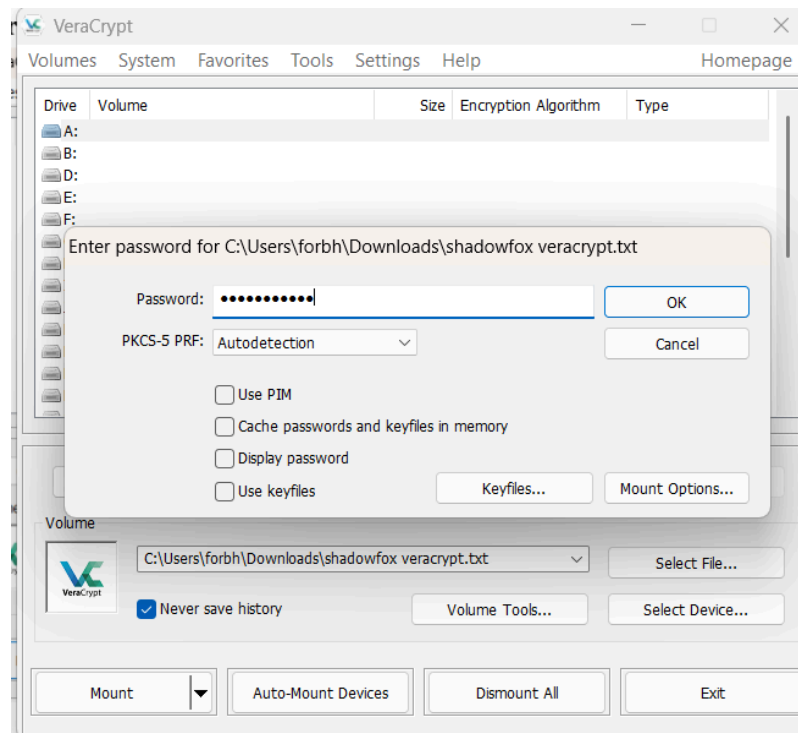


Fig 11: Password Authentication in VeraCrypt

Step 6: Retrieve the Secret Code

After the encrypted volume was successfully mounted, the contents of the mounted drive were accessed using File Explorer. The secret code stored inside the text file was located and retrieved.

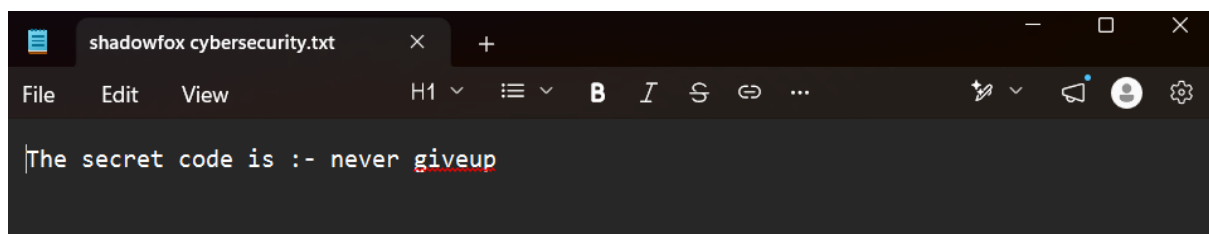


Fig 12: Secret Code Retrieved from the Mounted Volume

Analysis

The MD5 hash was successfully identified and decoded to recover the original password. The recovered password was then used to authenticate and mount the VeraCrypt encrypted container, providing access to the protected files and the secret code. The process demonstrated how weak or commonly used passwords can be recovered from insecure hashes and emphasized the importance of using strong passwords together with secure hashing algorithms to protect encrypted data.

Impact

Advantages

1. Demonstrates how encrypted data can be securely stored and accessed using the correct password.
2. Provides practical understanding of password recovery, hashing, and disk encryption concepts.

3. Highlights the importance of strong authentication in protecting sensitive information.

Disadvantages

1. Weak or commonly used passwords can be recovered using publicly available hash databases.
2. If the encryption password is compromised, unauthorized users can gain access to protected data.
3. Older hashing algorithms such as MD5 are vulnerable to lookup attacks and should not be used for securing passwords.

Mitigation Steps

1. Use strong, unique, and complex passwords for encrypted containers.
2. Replace MD5 with secure password hashing algorithms such as bcrypt, PBKDF2, or Argon2.
3. Apply a unique salt before hashing passwords to prevent hash lookup attacks.
4. Store encrypted containers only on trusted systems and restrict unauthorized access.
5. Regularly update encryption software to ensure the latest security improvements are applied.

Resources Used: Kali Linux, Hash-identifier, VeraCrypt, MD5 Encrypt & Decrypt, File Explorer, Windows Operating System, Encoded password file